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Teorema de diferenciación de Lebesgue en $\mathcal{L}^1$

Teorema 1 (de diferenciación de Lebesgue en $\mathcal{L}^1$). *Sea $f \in \mathcal{L}^1(\mathbb{R})$, entonces*

$$\lim_{h \to 0} \frac{1}{2h} \int_{x-h}^{x+h} f(t) \, \mathrm{d}t = f(x) \quad \textit{c.t.p. } x \in \mathbb{R}.$$

***Demostración*:** Por el `lem-diferenciacion-lebesgue-l1-imp-casi-toda-parte/`Lema 1, tenemos que $\forall n \in \mathbb{N}:$

$$A_{1/n} := \left\{ x \in \mathbb{R} : \limsup_{r \to 0^+} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) \, \mathrm{d}y - f(x) \right| > \frac{1}{n} \right\} \text{ tiene medida de Lebesgue cero.}$$

Tomamos $A = \bigcup_{n=1}^{\infty} A_{1/n}$, entonces $m(A) = 0$. Entonces, para $x \in A^{\mathsf{c}}$, tenemos que $\forall n \in \mathbb{N} : x \notin A_{1/n}$, es decir,

$$\forall n \in \mathbb{N} : \limsup_{r \to 0^+} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) \, \mathrm{d}y - f(x) \right| \leqslant \frac{1}{n} \implies \lim_{r \to 0^+} \frac{1}{2r} \int_{x-r}^{x+r} f(y) \, \mathrm{d}y = f(x). \qquad \blacksquare$$

Referenciado en

- `teo-diferenciacion-lebesgue`

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